



PRESENTED BY
Group 5

ZaraiLink - Trade Intelligence

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The problem in Trade

- **Trust Gap:** Foreign and Pakistani Traders cant reliably verify each other, so deals feel risky and opportunities are avoided.
- **Market Blindness:** Traders operate without clear visibility into real prices, volumes, or market trends.
- **Data Accesibility Gap:** Data exists but is fragmented, scattered across portals, spreadsheets and watsapp groups - no single complete view

How do we solve this problem?

ZaraiLink is a B2B trade intelligence platform connecting Pakistani businesses with global traders, enabling them to discover reliable trading partners and access structured trade data in a single system.

🔍 **Four ways to search:** HS Code, Category, Product Name, or plain English AI query.

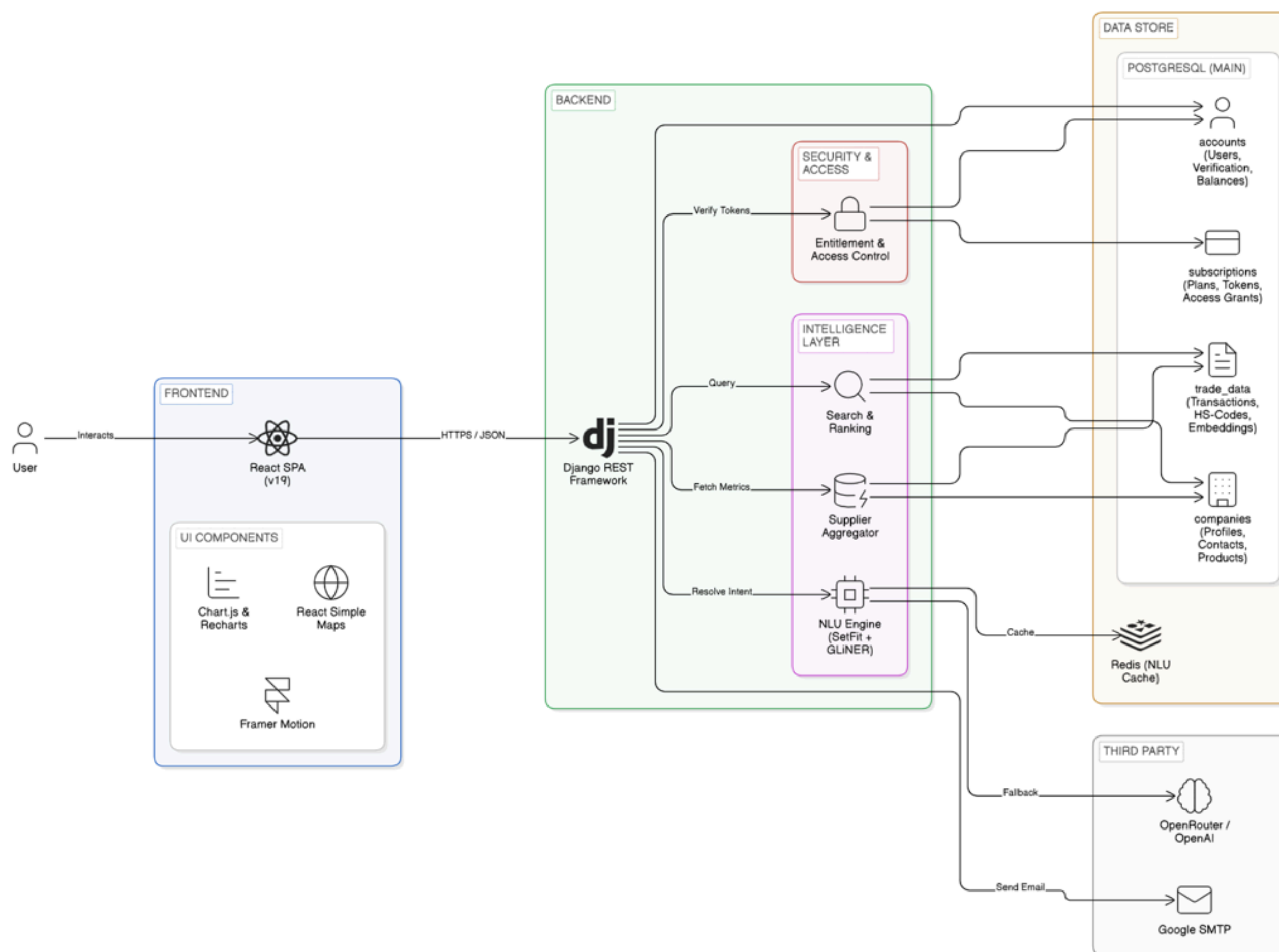


Deep market intelligence: prices, volumes, benchmarks, transaction history

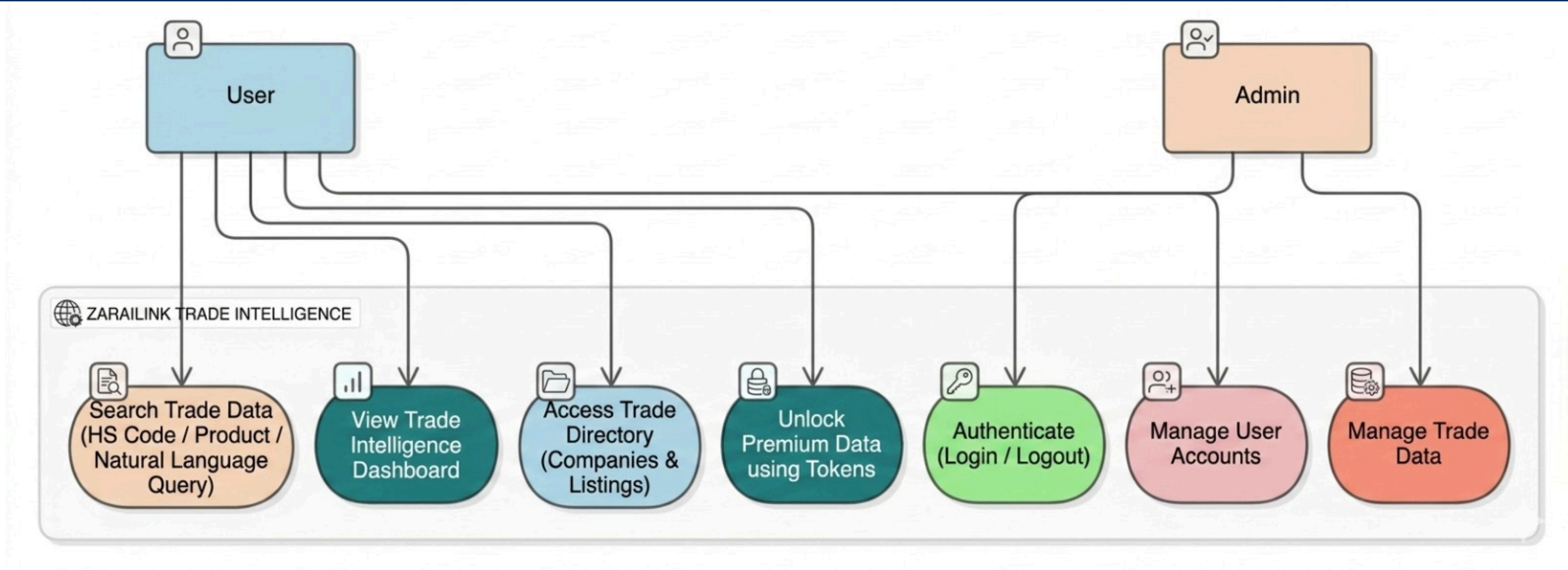


Verified partner discovery real buyers and suppliers, not listings

Requirements and Design - System Architecture



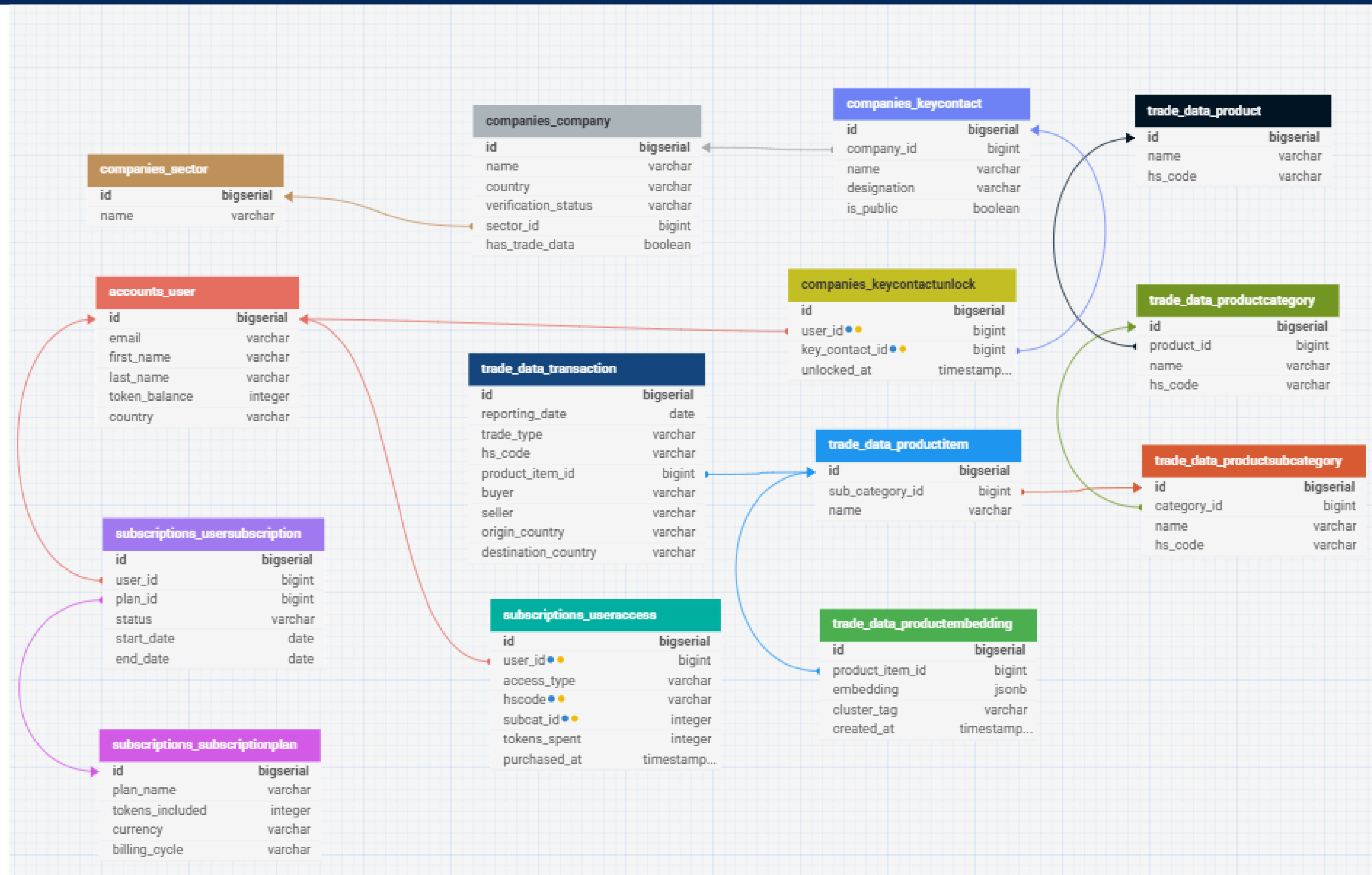
Requirements and Design - Use Case and Dataset



Dataset:

- 9,000+ trade records sourced from industry partner PAR
- 5,000 records independently cleaned, standardised, and HS code mapped by our team
- Covers sugar import and export transactions across 2024–2025
- Loaded into PostgreSQL using a hierarchical commodity structure (Product → Category → Sub-Category → Item)

Requirements and Design - ERD



ZaraiLink Demo

Functional Testing - Core Modules



Authentication System

- User registration, login, and logout
- Session management and protected routes
- Password reset and recovery
- Unauthorized access to protected pages



Access Control System

- Token-based paywall enforced across all modules
- Unauthorized access blocked at backend level
- Data unlock correctly updates user entitlement state
- Access remains consistent across search, dashboard, and profiles



Trade Intelligence System

- Trade data aggregated into structured market insights
- Supplier/buyer ranking and benchmarking
- Market dashboards and product analytics displayed aggregated trade data
- Deal detail views (company + transactions) function



Trade Directory

- Company search and filtering validated
- Company profile pages display structured data
- Contact masking and unlock mechanism tested

Non Functional Testing



Performance

- Search queries return results in under 3 seconds
- Deal detail and intelligence pages load quickly
- User profile and dashboard load instantly
- Autocomplete suggestions appear in real time



Security

- All premium data protected at server level
- Unauthenticated users cannot access trade data
- Paywall enforced in backend, not just frontend
- Contact details masked until explicitly unlocked
- Token transactions validated before processing

Reliability

- System remains stable under malformed, empty, and unexpected inputs
- Core modules handle invalid requests gracefully without crashing
- Search, authentication, and intelligence workflows remain stable under stress inputs
- Consistent behavior observed across all major system components

Usability

- Clear and intuitive navigation across all modules (search → results → details)
- Consistent UI behavior and layout across the platform
- Clear feedback for errors, empty states, and restricted access content
- Paywall and locked data clearly communicated to users

Search Engine - NLU Pipeline Evaluation		
Component	How it was Tested	Accuracy
Intent Classification (SetFit)	348 domain-specific trade queries, 80/20 split	89.4% accuracy
Product Extraction (GLiNER)	Clean queries, typos, synonyms, informal language, multi-word	Robust
Country Resolution (GLiNER + RapidFuzz + pycountry)	Typos, abbreviations, 249 countries	Robust
Price Extraction (Regex)	Explicit values (\$700/MT), implicit signals (cheap, bulk)	Robust
HS Code Search	Valid codes, invalid codes, partial prefixes, drill-down	Robust
Category & Product Name Search	Exact matches, partial matches, ambiguous inputs	Robust
Edge Case Coverage	60+ noisy, typo-heavy, garbage, contradictory queries	Stable
Query Latency	Timed across all tested queries end-to-end	200–500ms (target < 3s)

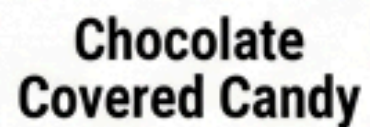
Limitations

- ✓ **Dataset Scope:** Platform currently covers sugar only. Broader commodity coverage would be needed for a production deployment serving Pakistan's full trade ecosystem.
- ✓ **Search Architecture:** Current structured database filtering limits relevance and speed at scale. Production systems use dense vector retrieval with dedicated reranking for significantly faster and more accurate results.
- ✓ **Ranking:** Current composite scoring could be replaced with a fully trained Learning-to-Rank model using real user engagement signals for more commercially relevant results.
- ✓ **Monetization:** Token system currently uses redeem codes as a dummy payment mechanism. Production deployment would require integration with a real payment gateway.

Future Work

-  **Urdu Language Support:** Extending the NLU pipeline to support Urdu and Roman Urdu queries would significantly improve accessibility for Pakistan's broader trader community..
-  **Mobile Application:** A dedicated mobile app would make trade discovery accessible to field-level traders and SMEs who primarily operate on mobile devices.
-  **Predictive Analytics:** Incorporating time-series modelling on historical trade records would enable price trend forecasting and demand prediction, giving users actionable foresight rather than just historical data

🔍 “I want to buy chocolate.”



Just looking
for a snack!

User

React SPA (v19)

/ JSON

Identify
HS-Code

NLU Intent Resolution (SetFit + GLiNER)

Search & Ranking Engine



Vector Store
(Embeddings)

Identify HS-Code

JSON lyper

INTELLIGENCE LAYER

NLU Inactionaitbnt Resolution (SetFit + GLINER)



Supplier Candiidating

Score	Supplier Reliability
1	1
2	2
3	3
4	4
5	5
6	6
7	7
8	8
9	9
10	10

Identify HS-Code

DATABASE

Transaction Records (HS-Code 1806)

Company Profiles

Product Hierarchy

Geopolitical Trade Maps

JSON Request

Score Supplier Reliability

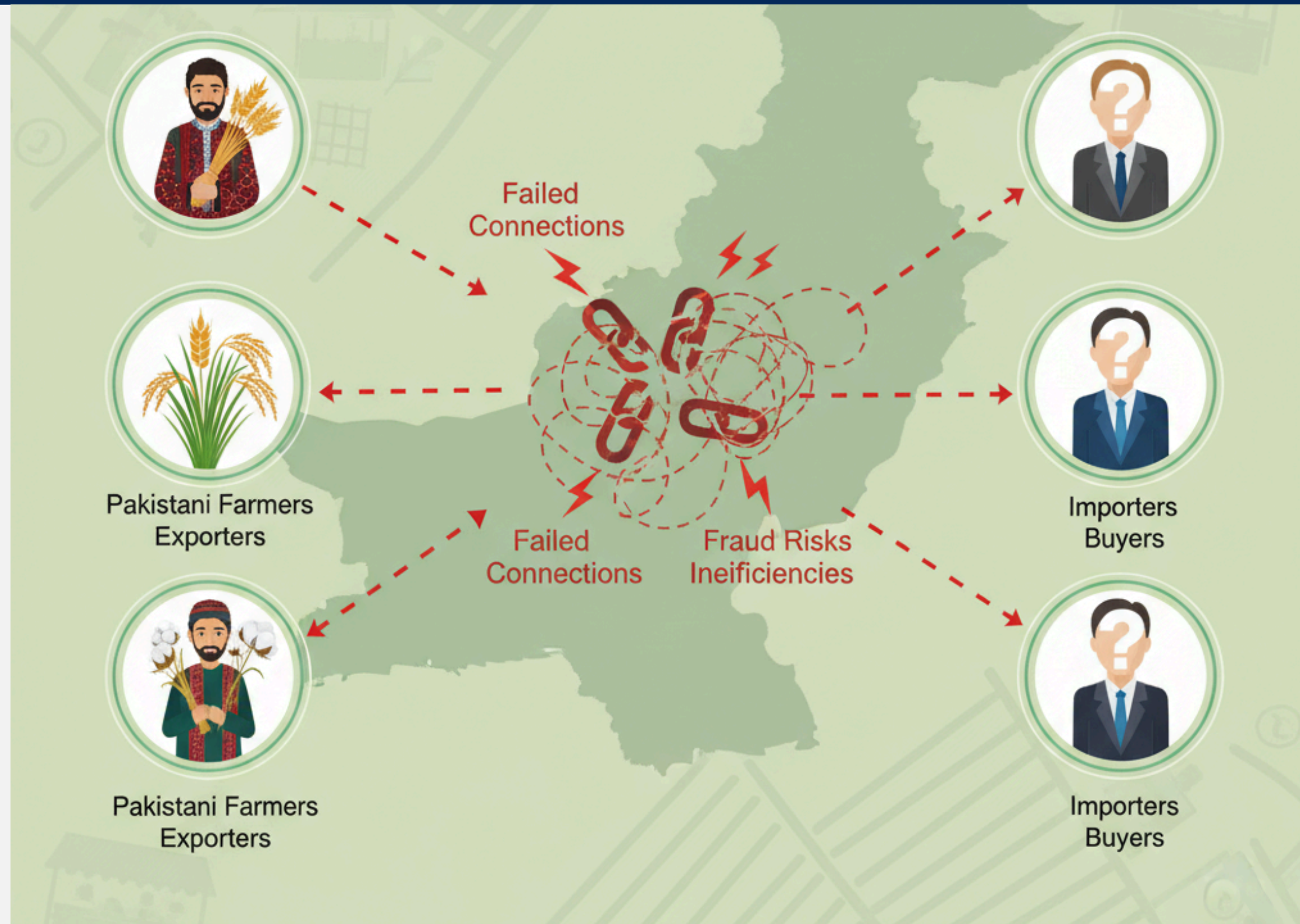
SECURITY & ENTITLEMENT

Verify Premium Token Access

Check Tokens

**Just found the
geopolitical trade data...
calculating supplier
reliability scores now!**

The Fragmented Agri-Trade Ecosystem in Pakistan



The problem in Agri-Trade



Trust gap: Exporters and buyers can't reliably verify each other, so deals feel risky and opportunities are avoided.



Market blindness: Low visibility into competitors, trade flows, and price trends leads to poor timing and lower margins.



Fragmented data: Trade information is scattered across portals, databases, spreadsheets, and chats—no single complete view.